

OSSP_SKILL_2

Step 1 – Open Terminal

Press Ctrl + Alt + T

Step 2 – Check GCC

```
gcc --version
```

```
srivarun@LAPTOP-TJEGMFM8:~$ gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

Step 3 – Install GCC if required

```
sudo apt update
```

```
sudo apt install build-essential -y
```

```
gcc --version
```

```
srivarun@LAPTOP-TJEGMFM8:~$ sudo apt update
[sudo] password for srivarun:
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
2 packages can be upgraded. Run 'apt list --upgradable' to see them.
srivarun@LAPTOP-TJEGMFM8:~$ sudo apt install build-essential -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
build-essential is already the newest version (12.10ubuntu1).
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amd-gpu-dri libglapi-mesa libllvm17t64 libwayland-server0 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
srivarun@LAPTOP-TJEGMFM8:~$ gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
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```

Step 4 – Create Project Folder

```
mkdir -p ~/ossp_practical3  
cd ~/ossp_practical3  
pwd
```

```
srivarun@LAPTOP-TJEGMFM8:~$ mkdir -p ~/ossp_practical3  
srivarun@LAPTOP-TJEGMFM8:~$ cd ~/ossp_practical3  
srivarun@LAPTOP-TJEGMFM8:~/ossp_practical3$ pwd  
/home/srivarun/ossp_practical3  
srivarun@LAPTOP-TJEGMFM8:~/ossp_practical3$ |
```

Step 5 – Create C File

```
nano myshell.c
```

```
srivarun@LAPTOP-TJEGMFM8:~/ossp_practical3$ nano myshell.c
```

Step 6 – Save the File

After pasting the program below into nano:

```
#include <stdio.h>  
#include <stdlib.h>  
#include <unistd.h>  
#include <termios.h>  
#include <sys/types.h>  
#include <sys/wait.h>  
#include <string.h>
```

```

#define MAX_INPUT 1024

void disableRawMode(struct termios *original)
{
    tcsetattr(STDIN_FILENO, TCSAFLUSH, original);
}

int main()
{
    struct termios original, raw;
    char input[MAX_INPUT];
    int index;
    char *args[100];

    while (1)
    {
        index = 0;

        printf("\nmyShell> ");
        fflush(stdout);

        if (tcgetattr(STDIN_FILENO, &original) == -1)
        {
            perror("tcgetattr");
            return 1;
        }

        raw = original;
        raw.c_lflag &= ~(ICANON | ECHO);

        if (tcsetattr(STDIN_FILENO, TCSAFLUSH, &raw) == -1)
        {
            perror("tcsetattr");
            return 1;
        }

        while (1)
        {
            char ch;

            if (read(STDIN_FILENO, &ch, 1) <= 0)
                break;

            if (ch == '\n' || ch == '\r')
            {
                input[index] = '\0';
                printf("\n");
                break;
            }

            if (ch == 127 || ch == '\b')
            {
                if (index > 0)
                {

```

```

        index--;
        printf("\b \b");
        fflush(stdout);
    }
    continue;
}

if (ch == 4)
{
    disableRawMode(&original);
    printf("\n");
    return 0;
}

if (index < MAX_INPUT - 1)
{
    input[index++] = ch;
    putchar(ch);
    fflush(stdout);
}
}

disableRawMode(&original);

if (index == 0)
    continue;

if (strcmp(input, "exit") == 0)
{
    printf("Exiting shell...\n");
    break;
}

int count = 0;
char *token = strtok(input, " ");

while (token != NULL && count < 99)
{
    args[count++] = token;
    token = strtok(NULL, " ");
}

args[count] = NULL;

if (count == 0)
    continue;

pid_t pid = fork();

if (pid < 0)
{
    perror("fork");
}
else if (pid == 0)
{

```

```

        execvp(args[0], args);
        perror("Command execution failed");
        exit(1);
    }
    else
    {
        waitpid(pid, NULL, 0);
    }
}

return 0;
}

```

- Press Ctrl + O
- Press Enter
- Press Ctrl + X

Step 7 – Check the File

Ls

```

srivarun@LAPTOP-TJEGMFM8:~/ossp_practical3$ nano myshell.c
srivarun@LAPTOP-TJEGMFM8:~/ossp_practical3$ ls
myshell.c

```

Step 8 – Compile

gcc myshell.c -o myshell

```

srivarun@LAPTOP-TJEGMFM8:~/ossp_practical3$ gcc myshell.c -o myshell
srivarun@LAPTOP-TJEGMFM8:~/ossp_practical3$ ls -l
total 24
-rwxr-xr-x 1 srivarun srivarun 16656 Aug 24 05:19 myshell
-rw-r--r-- 1 srivarun srivarun  2514 Aug 24 05:18 myshell.c
srivarun@LAPTOP-TJEGMFM8:~/ossp_practical3$ |

```

```
ls -l
```

Step 9 – Run the Shell

```
./myshell
```

```
srivarun@LAPTOP-TJEGMFM8:~/ossp_practical3$ ./myshell  
  
myShell> pwd  
/home/srivarun/ossp_practical3  
  
myShell> |
```

Step 10 – Test pwd

```
pwd
```

Step 11 – Test ls

```
Ls
```

```
myShell> ls  
myshell  myshell.c
```

Step 12 – Test ls -l

```
ls -l
```

```
myShell> ls -l  
total 24  
-rwxr-xr-x 1 anudeep anudeep 16656 Aug 24 05:19 myshell  
-rw-r--r-- 1 anudeep anudeep  2514 Aug 24 05:18 myshell.c
```

Step 13 – Test uname

```
Uname
```

```
myShell> uname  
Linux
```

Step 14 – Test date

Date

```
myShell> date  
Mon Aug 24 05:21:19 UTC 2026
```

Step 15 – Test multi-word command

echo Hello OSSP

```
myShell> echo Hello OSSP  
Hello OSSP
```

Step 16 – Test Backspace

Type: pwdd

```
myShell> pwdd  
Command execution failed: No such file or directory
```

Press Backspace once, then press Enter.

Step 17 – Test empty input

At myShell>, press Enter without typing anything.

```
myShell>
```

Step 18 – Test exit

Exit

```
myShell> exit  
Exiting shell...  
srivarun@LAPTOP-TJEGMFM8:~/ossp_practical3$ |
```

Step 19 – Test Ctrl+D

```
./myshell
```

```
srivarun@LAPTOP-TJEGMFM8:~/ossp_practical3$ ./myshell  
myShell>  
srivarun@LAPTOP-TJEGMFM8:~/ossp_practical3$ |
```

At myShell>, press Ctrl + D.